Lab Report: Image classification experiments

1 Introduction

In this report we compare several models on standard benchmarks. The dataset was split into 80 percent training data and the rest for testing. We trained each model for 50 epochs with a batch size of 32.

2 Results

Figure 1 shows the test accuracy of LSTM by dataset.

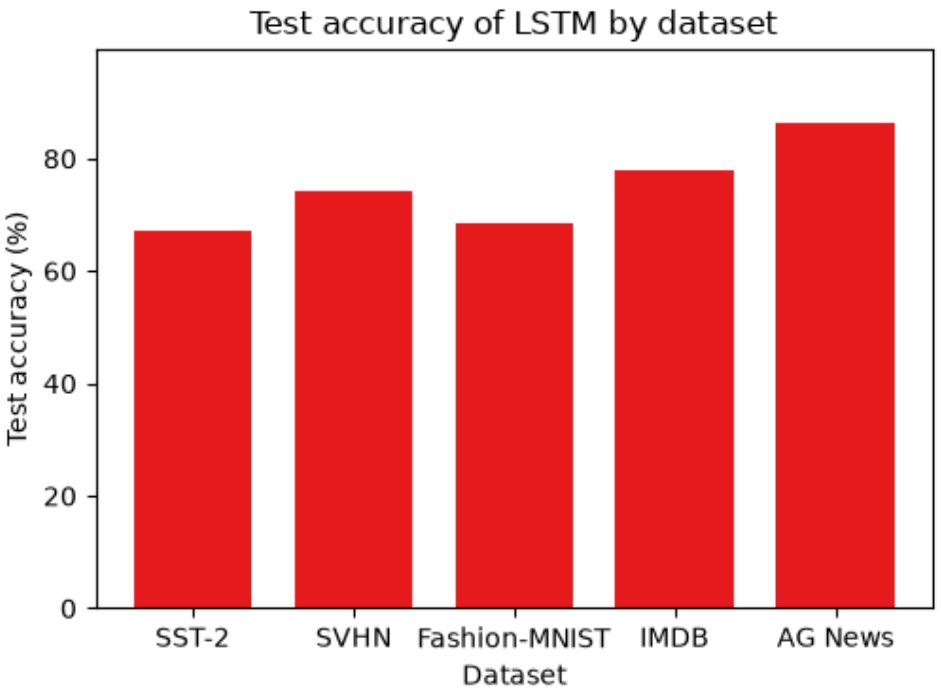


Figure 1: Test accuracy of LSTM by dataset for the models in this experiment.

LSTM achieves about 80% test accuracy on IMDB. LSTM scores 3.4 points higher on Fashion-MNIST than on SST-2. As shown in Figure 1, LSTM scores about 19 points higher on AG News than on SST-2. LSTM scores 23.0 points higher on AG News than on Fashion-MNIST.

Figure 2 shows the validation accuracy by dataset.

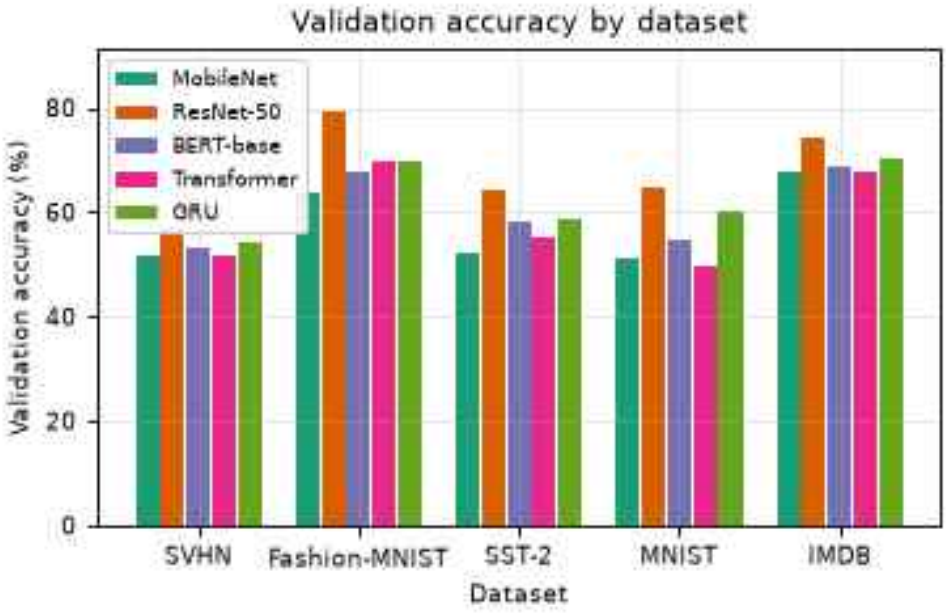


Figure 2: Validation accuracy by dataset for the models in this experiment.

GRU outperforms MobileNet by 8.8 percentage points on MNIST. MobileNet achieves 51.9% validation accuracy on SVHN. Transformer reaches 5.9% validation accuracy on SVHN. ResNet-50 outperforms BERT-base by about 4 percentage points on SVHN (Figure 2).

Figure 3 shows the test accuracy of ViT-S by dataset.

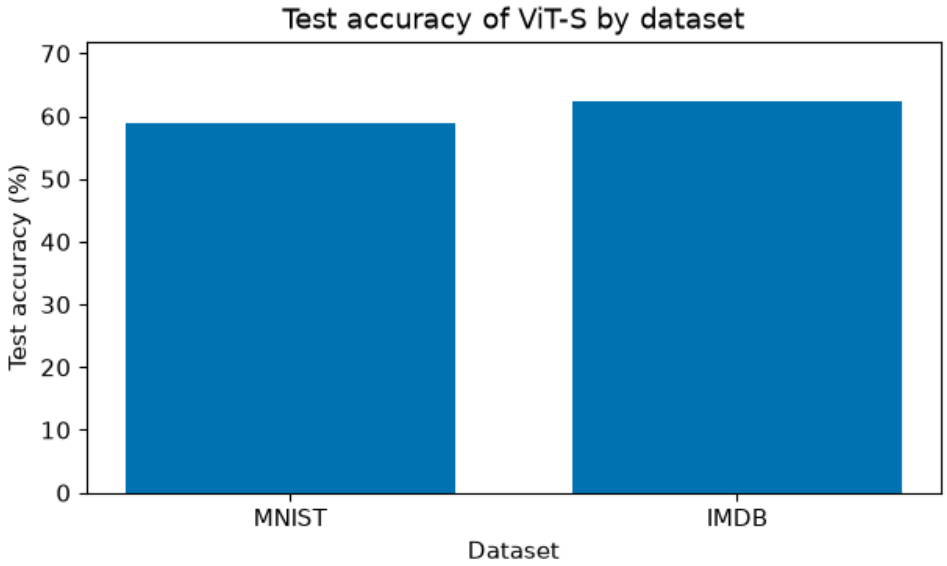


Figure 3: Test accuracy of ViT-S by dataset for the models in this experiment.

ViT-S scores 7.1 points higher on IMDB than on MNIST (Figure 3). On MNIST, ViT-S obtains a test accuracy of over 55%. On MNIST, ViT-S obtains a test accuracy of 62.5% (Figure 3).

3 Discussion

On IMDB, Random Forest obtains an accuracy of 74.8%. We note that the training was stable for all models.